

Nathan Hershey

Round Rock, TX | nathanhershey047@gmail.com | 937-326-3137
<https://www.linkedin.com/in/nathan-hershey-436209388>

Summary

Senior Full Stack Engineer with **8 years** of end-to-end product ownership, driving a high-throughput transaction platform and a multi-tenant analytics service. Regularly built on **Java, Spring Boot, Python, Docker, Kubernetes** while on call for kernel-level performance tuning. Delivered a 40% reduction in query latency for a flagship finance product, collaborating with data science, product, and operations partners. Managed the full software lifecycle from design through production, overseeing CI/CD pipelines, on-call incident response, and performance profiling across Linux and container environments. Mentored cross-functional teams to adopt best practices in API design and observability. Current focus is architecting intelligent caching and graph-based execution frameworks to accelerate compute workloads across the organization.

Core Skills

Languages and Frameworks: JavaScript, CSS

Backend Development: Java, Spring Boot, Python, REST, GraphQL, SQL

Container & Orchestration: Docker, Kubernetes, Helm, Terraform

Data & Performance: PostgreSQL, DuckDB, Redis, caching, profiling

Cloud & DevOps: AWS, CI/CD, Git, Jenkins, monitoring

Work Experience

FrontDoor, Senior Full Stack Engineer (09/2022 - Present)

- Re-engineered the core financial transaction engine using **Java, Spring Boot**, and **PostgreSQL**, cutting end-to-end processing time from 3 seconds to 1.2 seconds, improving customer satisfaction and reducing operational overhead for the payments team.
- Designed an intelligent **caching** layer with **Redis** that lowered repeat query latency by 45%, reducing compute cost for high-volume users across multiple microservices and enabled faster data retrieval for reporting dashboards.
- Implemented a **graph-based** dependency scheduler in **Python**, enabling parallel execution of multi-step workflows and increasing throughput by 30% for data-intensive pipelines, and reduced resource contention.
- Built end-to-end **REST** and **GraphQL** APIs for the collaborative notebook interface, supporting on-call debugging and kernel-level profiling that allowed external partners to embed compute cells and monitor execution metrics in real time.
- Integrated **DuckDB** for smooth data interchange between **Python** scripts and SQL queries, expanding analytic capabilities for internal data scientists and reducing ETL latency by half for exploratory notebooks.
- Established comprehensive monitoring with **Prometheus** and **Grafana**, cutting mean time to detection by 50% and providing actionable dashboards for performance tuning across all services.

ZenBusiness, Full Stack Developer (01/2021 - 11/2022)

- Created a multi-tenant SaaS platform on **Python, Django**, and **PostgreSQL**, handling over 200 k monthly active users and delivering secure isolation between client data sets while meeting PCI compliance requirements for financial transactions.
- Added **Docker-based CI/CD** pipelines that reduced deployment lead time from hours to minutes, enabling rapid feature iteration across cross-functional squads and decreasing rollback frequency by 20% through automated testing and container health checks.
- Developed a real-time notification service using **WebSockets** and **real-time** techniques, improving user engagement metrics by 20% and providing instant feedback for critical workflow events while also reducing email overload for support teams.

- Optimized database indexes and query plans in **PostgreSQL** and **SQL**, decreasing average report generation time by 35% and freeing resources for concurrent analytics workloads while maintaining strict SLA compliance across all tenant accounts.
- Collaborated with product and operations teams to define API contracts, delivering stable **REST** and **API** endpoints for third-party integrations and achieving a 99.9% uptime SLA through rigorous versioning and automated contract testing.

Northstar Software Development, Full Stack Engineer (11/2020 - 06/2021)

- Delivered an applicant tracking system using **Node.js**, **Express**, and **React**, supporting 5 k concurrent users and streamlining recruiter workflows across multiple regions while ensuring GDPR compliance for candidate data.
- Implemented role-based access control with **JWT** and **security** enhancements, reducing unauthorized access incidents by 40% and improving audit traceability for compliance teams.
- Containerized the application with **Docker** and **Kubernetes**, enabling rapid scaling during peak recruitment periods and cutting infrastructure provisioning time by 70% while maintaining consistent environment configurations across dev, test, and prod.
- Automated infrastructure provisioning via **Terraform** and **infrastructure** scripts, cutting environment setup time by 70% and improving disaster-recovery readiness, while reducing manual errors by 35%.
- Monitored performance with **New Relic** and **monitoring**, identifying bottlenecks that were resolved to improve page load speed by 25% and enhance overall user experience across both desktop and mobile clients, supporting a 15% increase in conversion rates.

Arbisoft, Software Engineer (10/2018 - 08/2020)

- Built the FixAlert monitoring suite using **Java**, **Spring MVC**, and **MySQL**, delivering alerts for over 1 M devices and providing a unified dashboard for operations teams while supporting multi-tenant configurations for enterprise customers.
- Created **REST** services that processed 10 k requests per minute with sub-second latency, enabling real-time device health monitoring and integrating with third-party ticketing systems for automated incident escalation across global data centers.
- Integrated **Kafka** and **streaming** for event streaming, enabling real-time analytics for client dashboards and reducing data propagation delays by 50% while ensuring exactly-once delivery semantics across distributed services and simplifying downstream processing pipelines.
- Wrote unit and integration tests with **JUnit** and **testing**, raising code coverage to 85% and catching regressions before production releases, which decreased post-release defect rate by 30% and improved developer confidence across the team.
- Participated in on-call rotations, troubleshooting **kernel-level memory** issues that reduced crash incidents by 15% and improving overall system stability while documenting remediation steps for future reference and knowledge sharing.

Education

Bachelor of Science in Computer Science

University of North Texas | 2014 - 2018